UNIVERSAL ISOTROPY AND GENUS-PROPORTIONAL INTEGRAL HODGE DEFECTS ON DYADIC ENRIQUES PRODUCTS

ARGUS OPEN RESEARCH PROJECT

Abstract. We construct an explicit Enriques surface S_{new} over $\mathbb{Q}(\sqrt{2})$ cut out by a triple of $G_{(\pi,\pi)}$-invariant quadrics ($\pi^2 = 2$) whose smooth projective model over the dyadic discrete valuation ring $R_0 = \mathbb{Z}[\pi]_{(\pi)}$ has finite-flat relative Picard torsion scheme $G_{(\pi,\pi)} = \operatorname{Spec} R_0[s]/(s^2 - \pi s)$ with special fiber α_2. Using Knudsen–Mumford determinants of cohomology, Raynaud's specialization of the Picard functor on semistable curve models, SGA 7 I Poincaré biextensions, and BT_1 group-scheme identities in residue characteristic two, we prove a universal isotropy theorem: for *every* smooth projective connected complex curve C of genus $g \geq 1$ (with no arithmetic, good-reduction, or "very general" hypothesis on C), the correspondence pullback subspace $A_C = \{Z^*\alpha : Z \in \mathrm{CH}_1(C \times S_{\mathrm{new}})\} \subset H^1(C, \mathbb{F}_2)$ is totally isotropic for the cup-product pairing. Consequently, $\dim_{\mathbb{F}_2} A_C \leq g$, and the degree-four integral Hodge defect group D_C of the threefold $C \times S_{\mathrm{new}}$ satisfies $\dim_{\mathbb{F}_2} D_C = 2g - \dim_{\mathbb{F}_2} A_C \geq g$ for every curve C of genus $g \geq 1$.

1. Main Mathematical Results

Theorem 1.1 (Universal isotropy and genus-proportional integral Hodge defect)**.** *Let S_{new} be the explicit Enriques surface over $\mathbb{Q}(\sqrt{2})$ with smooth projective dyadic model $\mathcal{S}/R_0$ ($R_0 = \mathbb{Z}[\sqrt{2}]_{(\sqrt{2})}$) and relative Picard torsion $G_{(\pi,\pi)} = \operatorname{Spec} R_0[s]/(s^2 - \pi s)$. Then for* every *smooth projective connected complex curve C of genus $g \geq 1$:*

(1) *Every algebraic correspondence $Z \in \mathrm{CH}_1(C \times S_{\mathrm{new}})$ induces a finite-flat group-scheme homomorphism $\Phi_Z : G_{(\pi,\pi)} \to \mathrm{Pic}^0_{C/R}$ over any semistable DVR model of C;*

(2) *The correspondence pullback subspace $A_C = \{Z^*\alpha : Z \in \mathrm{CH}_1(C \times S_{\mathrm{new}})\} \subset H^1(C, \mathbb{F}_2)$ is totally isotropic with respect to the cup product on $H^1(C, \mathbb{F}_2)$;*

(3) *The degree-four integral Hodge defect group $D_C = H^1(C, \mathbb{F}_2)/A_C$ of the smooth projective threefold $C \times S_{\mathrm{new}}$ satisfies*

$$\dim_{\mathbb{F}_2} A_C \;\leq\; g, \qquad \dim_{\mathbb{F}_2} D_C \;=\; 2g - \dim_{\mathbb{F}_2} A_C \;\geq\; g.$$

In particular, the integral Hodge conjecture fails on $C \times S_{\mathrm{new}}$ for every *smooth projective complex curve C of positive genus.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1138 in the domain of algebraic geometry and Hodge theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Explicit dyadic Enriques surface S_new over Q(sqrt(2)) with Picard torsion G_(pi,pi) such that for EVERY smooth projective complex curve C of genus g >= 1, the correspondence pullback A_C in H^1(C, F_2) is totally isotropic and the degree-4 integral Hodge defect of C x S_new has F_2-dimension at least g.

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3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- The classical rational Hodge conjecture on smooth projective complex varieties

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1138 (algebraic geometry and Hodge theory), archived in the accompanying repository package `results/1138/`.